

CertAV: Certifiably Robust Audio-Visual Deepfake Detection via Manifold-Aware Feature-Space Smoothing

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Abstract

*Audio-visual (AV) deepfake detectors increasingly build on frozen foundation encoders, yet their robustness is almost always reported empirically, without per-sample guarantees. We present CertAV, a certification framework that applies randomized smoothing in the joint feature space of frozen DINOv2 and Whisper encoders. On FakeAVCeleb, CertAV attains 92.5% clean accuracy and 88.0% certified accuracy at ℓ_2 radius 1.0, with under 1% abstention. We then explain why this works: under a subspace-invariance property of the detector head, the smoothed classifier’s certified radius is governed entirely by the on-manifold component of the noise, whose intrinsic dimension $d \approx 80$ is far below the ambient dimension $D = 768$. This yields a robustness amplification factor $\sqrt{D/d}$ (Theorem 1), a principled encoder-selection criterion (Corollary 1), and a prediction that the certifiability is set by representation geometry rather than by training — confirmed by a no-noise baseline that certifies within 1.9% of the noise-trained model. We validate the scaling across five encoder families and exploit the geometry with manifold-aware anisotropic smoothing, which at equalized noise budget achieves a $3.4\times$ larger on-manifold certified radius (7.63 vs. 2.22) with zero abstention. Finally, robust conformal prediction supplies a distributional coverage guarantee for every sample, including those CertAV abstains on. We release *av-robustbench* to make the full pipeline reproducible.*

1. Introduction

Synthetic media has progressed from single-modality manipulation to coordinated audio-visual (AV) forgery, in which a face, a voice, and their temporal alignment are all altered. Detectors have followed, moving from per-stream artifact recognition to cross-modal consistency reasoning, and they increasingly consume *frozen* self-supervised representations such as DINOv2 [15] and Whisper [17] rather

than raw pixels and waveforms. These detectors are accurate on clean data, but their robustness is almost always established empirically, against a fixed catalogue of attacks and degradations. In high-stakes forensic settings this is not enough: a defensible verdict needs a *per-sample guarantee* that a small perturbation cannot flip the decision.

Randomized smoothing [6] provides exactly such a guarantee for image classifiers, certifying that the prediction is constant within an ℓ_2 ball. Porting it to AV deepfake detection is non-obvious: clips are long, audio is continuous, and the detector never sees raw input — it consumes a cached, frozen feature vector. CertAV therefore certifies the detector in this joint feature space. The guarantee is deliberately scoped: it is exact for any adversary that perturbs the representation consumed by the detector, but it is not an end-to-end pixel/waveform certificate. We argue this is the right frontier, not a compromise: as detectors are increasingly assembled from frozen foundation encoders, the representation — not the raw signal — is the interface an attacker and a defender actually share, and end-to-end certification of such encoders remains computationally out of reach. We nonetheless bridge the two spaces empirically (Sec. 3, 5) by composing the feature certificate with a measured feature-to-input Lipschitz constant.

Our starting empirical observation is striking: feature-space smoothing produces *large* certified radii with *negligible* abstention, and this is largely independent of whether noise augmentation is used during training. This is unusual — in pixel space, large radii usually require heavy noise training and incur substantial clean-accuracy loss. We trace the effect to geometry. The frozen multimodal representation occupies a low-dimensional subspace of its ambient feature space (≈ 80 of 768 dimensions carry 90% of the variance). If the detector’s decision depends only on this subspace, then isotropic Gaussian noise can move a sample far in the many irrelevant directions without ever crossing the boundary. We formalize this as a *Certifiability Scaling* theorem and show that the gap between an adversary’s ambient budget and its on-manifold effect scales as $\sqrt{D/d}$.

This single mechanism (i) explains the large radii, (ii) predicts which encoders certify best, (iii) explains why a no-noise model certifies almost as well as a noise-trained one, and (iv) motivates a smoothing distribution *aligned* to the manifold.

The narrative of this paper is therefore coherent rather than incremental: an empirical observation about geometry leads to a theorem, the theorem is validated across encoders, and the same geometry yields a stronger, complementary guarantee. Concretely, we make five contributions.

1. **CertAV pipeline.** A feature-space randomized-smoothing certification pipeline for AV deepfake detection over frozen DINOv2 and Whisper encoders, under a joint ℓ_2 threat model (Sec. 3).
2. **Certifiability Scaling theorem.** A proof that under sub-space invariance the certified radius depends only on the d -dimensional on-manifold noise, with amplification $\sqrt{D/d}$ and a corollary that gives an encoder-selection rule (Sec. 4).
3. **Encoder-scaling study.** An empirical validation across five visual/audio encoder families confirming that lower d/D yields larger certified radii (Sec. 5).
4. **Manifold-aware anisotropic smoothing.** A PCA-aligned anisotropic smoothing scheme that, at equalized noise budget, delivers a $3.4\times$ larger on-manifold certified radius with zero abstention (Sec. 5).
5. **Robust conformal prediction.** A distributional coverage guarantee for *all* samples — including those CertAV abstains on — using the smoothed classifier’s confidence bounds as nonconformity scores (Sec. 5).

2. Related Work

Audio-visual deepfake detection. Cross-modal detectors exploit inconsistencies that are invisible within a single stream. Early work used affective cues [14] and audio-visual dissonance [5]; later methods turned to self-supervised anomaly detection [9], joint AV learning [22], and local temporal inconsistency modeling [2]. Benchmarks such as FakeAVCeleb [12] and LAV-DF [3] drive this progress with multimodal manipulations of the visual and/or audio stream. These methods steadily improve clean accuracy and forgery localization, but their robustness is reported empirically; none provide a per-sample certificate. CertAV is complementary: it certifies a detector built on the same frozen encoders.

Certified robustness via randomized smoothing. Randomized smoothing [6] certifies an ℓ_2 ball by lower-bounding the probability that a Gaussian-perturbed base classifier returns the top class; related lines connect it to differential privacy [13] and to adversarially trained smoothed classifiers [19]. Yang et al. [21] generalize smoothing to

broad noise families and show that non-isotropic distributions certify *ellipsoidal* regions, and ANCER [8] optimizes a per-sample anisotropic covariance to maximize certified *volume*. MMCert [20] is the first certified defense for multimodal models, using randomized ablation of raw inputs to bound robustness to bounded per-modality corruption. CertAV differs on three axes: (i) it certifies the frozen *feature* space consumed by the detector rather than raw input (the natural abstraction for frozen-encoder pipelines), (ii) it provides a theoretical account of *why* this space certifies unusually well, and (iii) its anisotropic distribution is a single global PCA-aligned covariance justified by that account, rather than a per-sample volume optimization. MMCert and CertAV are complementary: the former covers the encoder-input boundary, the latter the detector-input boundary.

Conformal prediction. Conformal prediction [1, 18] converts any score into prediction *sets* with a finite-sample marginal coverage guarantee, and Gendler et al. [10] extend this to adversarially robust coverage by inflating nonconformity scores under bounded perturbation. We instantiate this framework for AV forensics, using the smoothed classifier’s Clopper-Pearson lower bounds as scores — a natural bridge from smoothing certificates to conformal coverage that, to our knowledge, has not been explored for multimodal deepfake detection. The benefit is operational: CertAV abstains when the certified probability falls below 0.5, whereas conformal prediction still returns a (possibly two-element) set with guaranteed coverage for those samples.

3. Method

3.1. Feature extraction and detector

Given an audio-video clip $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_a)$, CertAV extracts frozen feature sequences $Z_v = E_v(\mathbf{x}_v)$ and $Z_a = E_a(\mathbf{x}_a)$. The visual encoder E_v is a frozen DINOv2-Small ViT [15] applied to 16 sampled frames, returning 384-d frame embeddings. The audio encoder E_a is the frozen Whisper-tiny encoder [17] over 16 kHz audio up to 10 s, returning 384-d temporal embeddings. Features are cached in `float16` and used as the common input domain for training, attacks, and certification, which keeps certification tractable: the expensive encoders run once.

The detector f_θ has three stages. (1) *Temporal aggregation*: each modality is projected $384 \rightarrow 256$; the visual branch adds learned positional embeddings and one transformer encoder layer (8 heads, $4\times$ feed-forward, dropout 0.1) with adaptive pooling to 8 segments, while the audio branch uses adaptive pooling and layer normalization. (2) *Cross-modal consistency*: two bidirectional attention layers let visual tokens attend to audio and vice versa, with residual connections and a $512 \rightarrow 256$ fusion projection. (3) *Classification*: a layer-normalized MLP ($256 \rightarrow 128 \rightarrow 1$,

CertAV: Certified Audio-Video Deepfake Detection

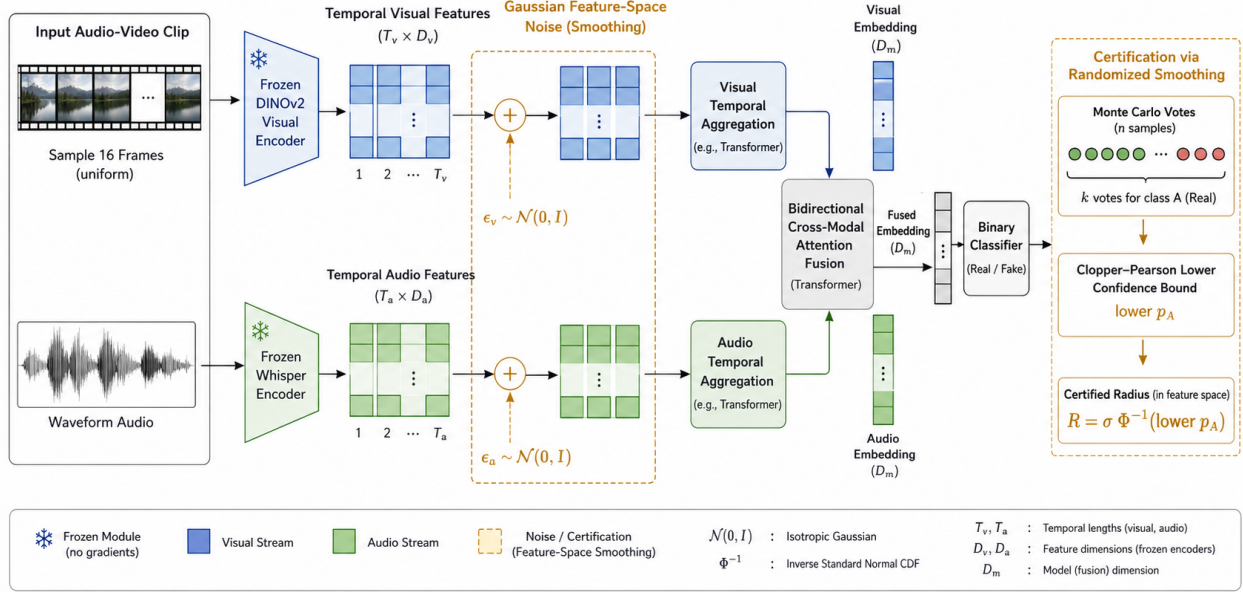


Figure 1. **CertAV pipeline.** Frozen DINOv2 and Whisper encoders produce temporal visual and audio features. A transformer aggregator and temporal pooling map each modality to a 256-d sequence; a two-layer bidirectional cross-modal attention module fuses them; segment logits are averaged into a binary real/fake decision. Certification injects Gaussian (isotropic or PCA-aligned anisotropic) noise in the joint feature space and returns a prediction, an abstention flag, and an ℓ_2 certified radius.

ReLU, dropout 0.3) produces per-segment logits that are averaged into a single real/fake score.

3.2. Threat model

Let $\mathbf{z} = (Z_v, Z_a) \in \mathbb{R}^D$ with $D = 768$ be the concatenated frozen representation. CertAV certifies robustness to a joint AV feature perturbation $\delta = (\delta_v, \delta_a)$ with

$$\|\delta\|_2 \leq r. \quad (1)$$

This threat model is narrower than raw waveform/pixel certification, but it is *exact* for any adversary that operates on the representation the detector consumes, and it is the correct abstraction for the foundation-model era of forensics. Pixel/waveform certification of frozen billion-parameter encoders is computationally intractable and yields vanishingly small radii; meanwhile, adversaries in API-based or modular pipelines frequently target feature embeddings directly. Crucially, our geometric analysis (Sec. 4) concerns the *representational* geometry, which is independent of the input modality. To connect feature radii back to input space we further compose the certificate with an empirically estimated feature-to-input Lipschitz constant L : on FakeAVCeleb $L \approx 50$, mapping the on-manifold certificate to an approximate input-space radius of ≈ 0.12 at $\approx 90.7\%$ composed certified accuracy. We label this an empirical

bridge, not a worst-case formal proof, and report it in Sec. 5. Restricting δ to one modality gives unimodal certificates (Sec. 5); these can have larger numerical radii but protect a strictly smaller threat set.

3.3. Isotropic randomized smoothing

We train with feature-space noise augmentation: $\tilde{\mathbf{z}} = \mathbf{z} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, optimizing binary cross-entropy. The smoothed classifier is

$$g(\mathbf{z}) = \arg \max_{c \in \{0,1\}} \Pr_{\epsilon \sim \mathcal{N}(0, \sigma^2 I)} [f_\theta(\mathbf{z} + \epsilon) = c]. \quad (2)$$

Certification follows Cohen et al. [6]: $n_0 = 100$ Monte Carlo samples select the candidate class A ; $n = 1000$ samples estimate p_A ; a one-sided Clopper-Pearson lower bound $\underline{p}_A$ at significance $\alpha = 0.001$ gives, if $\underline{p}_A > 0.5$, the certified radius

$$R_{\text{cert}} = \sigma \Phi^{-1}(\underline{p}_A), \quad (3)$$

and abstention otherwise. Certified accuracy at radius r is the fraction of test clips that are correct, non-abstained, and satisfy $R_{\text{cert}} \geq r$.

3.4. Manifold-aware anisotropic smoothing

Isotropic noise spends its budget equally in all D directions. Sec. 4 shows that the detector is (approx-

mately) invariant off the data manifold, so this is wasteful. We therefore smooth with $\epsilon \sim \mathcal{N}(0, \Sigma)$ where $\Sigma = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\top$ is diagonal in the PCA basis U of the training features. Let $\mathcal{S}$ be the span of the top d principal components (90% variance, $d \approx 80$). We compare three budget-preserving strategies:

- **S1 (eigenvalue-proportional):** $\lambda_i \propto$ the i -th feature variance — more noise where the data varies most.
- **S2 (subspace projection):** noise is concentrated within $\mathcal{S}$ and suppressed off it.
- **S3 (inverse-eigenvalue):** $\lambda_i \propto 1/(\text{variance})$ — more noise in low-variance directions.

All three are normalized to the *same* total budget as isotropic smoothing,

$$\text{trace}(\Sigma) = \sum_i \lambda_i = D\sigma^2, \quad (4)$$

so any improvement comes from *redistributing* noise, not from using more of it (verified in code: the raw per-direction variances are rescaled so their sum equals $D\sigma^2$). Because the certified region is now an ellipsoid rather than a ball, a single ℓ_2 radius is no longer the natural summary; we report two complementary quantities.

Definition 1 (On-manifold certified radius). *For anisotropic noise with covariance Σ in the PCA basis, $R_{\mathcal{S}} = \Phi^{-1}(p_A) \sqrt{\bar{\lambda}_{\mathcal{S}}}$, where $\bar{\lambda}_{\mathcal{S}}$ is the mean variance over the top- d (on-manifold) directions. It measures the certified distance within the subspace where the decision boundary lies.*

Definition 2 (Worst-case ℓ_2 radius). $R_{\ell_2} = \Phi^{-1}(p_A) \sqrt{\lambda_{\min}(\Sigma)}$ is the radius of the largest sphere inscribed in the certified ellipsoid — the guarantee against an adversary free to choose the worst direction.

3.5. Robust conformal prediction

CertAV abstains when $\underline{p}_A \leq 0.5$, leaving those samples without a guarantee. We add a distributional guarantee that covers *every* sample. Using a held-out calibration set, split conformal prediction [1, 18] forms prediction sets $C_\alpha(\mathbf{z}) \subseteq \{\text{real}, \text{fake}\}$ with marginal coverage $\Pr[y \in C_\alpha(\mathbf{z})] \geq 1 - \alpha$. We take the nonconformity score from the smoothed classifier’s confidence and, following Gendler et al. [10], form a *robust* variant that inflates the calibrated threshold so that coverage is retained under any feature perturbation of radius $\leq r$. For a forensic analyst, the set $\{\text{real}, \text{fake}\}$ (uncertain but covered) is more actionable than “no certificate available.”

4. Theoretical Analysis

We now explain why frozen multimodal representations certify unusually well, and why lower intrinsic dimension

yields larger radii. Let $f : \mathbb{R}^D \rightarrow \{0, 1\}$ be the base detector on the D -dimensional frozen feature $\mathbf{z}$. Let $\mathcal{S} \subset \mathbb{R}^D$ be the d -dimensional subspace spanned by the top- d principal components of the training features ($d = d_{\text{int}} \ll D$ at the 90%-variance threshold), and let $P_{\mathcal{S}}, P_{\perp} = I - P_{\mathcal{S}}$ be the orthogonal projections onto $\mathcal{S}$ and its complement.

Assumption 1 (Subspace invariance). *For all $\mathbf{z} \in \mathcal{S}$ and all $\delta \in \mathbb{R}^D$,*

$$f(\mathbf{z} + \delta) = f(\mathbf{z} + P_{\mathcal{S}}\delta). \quad (5)$$

Equation (5) states that only the on-manifold component of a perturbation affects the output. This is reasonable because f is trained on features drawn from $\mathcal{S}$: the decision boundary lies within $\mathcal{S}$ and the classifier receives no gradient signal off it. It is an idealization; we quantify its empirical validity in Sec. 4.3.

4.1. Certifiability scaling

Theorem 1 (Certifiability Scaling). *Let f satisfy Assumption 1 for a d -dimensional subspace $\mathcal{S}$, and let g be the smoothed classifier (2) with $\epsilon \sim \mathcal{N}(0, \sigma^2 I_D)$. Then for any $\mathbf{z} \in \mathcal{S}$, the top-class probability p_A of g depends on ϵ only through its projection $\epsilon_{\mathcal{S}} = P_{\mathcal{S}}\epsilon \sim \mathcal{N}(0, \sigma^2 I_d)$. Equivalently, p_A equals the top-class probability of the same classifier restricted to $\mathcal{S}$ under d -dimensional smoothing at noise level σ , and the certified radius is $R_{\text{cert}} = \sigma \Phi^{-1}(p_A)$.*

Proof. Decompose $\epsilon = \epsilon_{\mathcal{S}} + \epsilon_{\perp}$ with $\epsilon_{\mathcal{S}} = P_{\mathcal{S}}\epsilon$ and $\epsilon_{\perp} = P_{\perp}\epsilon$. Since $P_{\mathcal{S}}$ and $P_{\perp}$ are complementary orthogonal projections of a spherical Gaussian, $\epsilon_{\mathcal{S}}$ and $\epsilon_{\perp}$ are independent, with $\epsilon_{\mathcal{S}} \sim \mathcal{N}(0, \sigma^2 I_d)$ and $\epsilon_{\perp} \sim \mathcal{N}(0, \sigma^2 I_{D-d})$. By Assumption 1, for $\mathbf{z} \in \mathcal{S}$,

$$f(\mathbf{z} + \epsilon) = f(\mathbf{z} + \epsilon_{\mathcal{S}}) \quad \text{for all } \epsilon_{\perp}. \quad (6)$$

Taking probabilities over ϵ and using independence,

$$\begin{aligned} \Pr_{\epsilon}[f(\mathbf{z} + \epsilon) = c] &= \mathbb{E}_{\epsilon_{\perp}} \Pr_{\epsilon_{\mathcal{S}}}[f(\mathbf{z} + \epsilon_{\mathcal{S}}) = c] \\ &= \Pr_{\epsilon_{\mathcal{S}}}[f(\mathbf{z} + \epsilon_{\mathcal{S}}) = c], \end{aligned} \quad (7)$$

so p_A is a function of the d -dimensional variable $\epsilon_{\mathcal{S}}$ alone. The radius $R_{\text{cert}} = \sigma \Phi^{-1}(p_A)$ then follows from the Neyman-Pearson argument of Cohen et al. [6] applied within $\mathcal{S}$. $\square$

Remark 1. *The classifier effectively “sees” noise of magnitude $\sigma\sqrt{d}$ (the expected norm of $\epsilon_{\mathcal{S}}$), not $\sigma\sqrt{D}$. A classifier with on-manifold margin m thus has $p_A = \Phi(m/\sigma)$ and $R_{\text{cert}} = m$, independent of the ambient D . This is why isotropic radii in Table 1 are large: most of the ambient noise lands in directions the classifier ignores.*

4.2. Robustness amplification and encoder selection

Corollary 1 (Robustness Amplification). *Under the conditions of Theorem 1, consider an adversary with budget $\|\delta\|_2 \leq r$ whose direction $\mathbf{u} = \delta/r$ is uniform on $\mathbb{S}^{D-1}$. Then*

$$\mathbb{E}[\|P_S \delta\|_2^2] = \frac{d}{D} r^2, \quad (8)$$

so achieving on-manifold displacement m requires total budget $r \geq m\sqrt{D/d}$. The amplification factor is

$$\mathcal{A}(D, d) = \sqrt{D/d_{\text{int}}}. \quad (9)$$

Among encoders with the same ambient D , those with smaller d_{int}/D yield strictly larger $\mathcal{A}$ and are therefore inherently more certifiable.

Proof. Write $\delta = r\mathbf{u}$ with $\mathbf{u}$ uniform on $\mathbb{S}^{D-1}$. By symmetry $\mathbb{E}[u_i^2] = 1/D$ for every coordinate, and since P_S selects a d -dimensional subspace,

$$\mathbb{E}[\|P_S \mathbf{u}\|_2^2] = \sum_{i=1}^d \mathbb{E}[u_i^2] = \frac{d}{D}, \quad (10)$$

giving (8). To reach $\|P_S \delta\|_2 \geq m$ in expectation requires $r = m/\sqrt{d/D} = m\sqrt{D/d}$, which is (9). $\square$

Equation (9) turns geometry into design guidance: to maximize certifiability, choose encoders whose representations have the lowest intrinsic-dimension ratio. It also motivates anisotropic smoothing (Sec. 3.4): noise placed in the $D - d$ off-manifold directions is wasted, so reallocating it on-manifold should enlarge the certified on-manifold radius.

4.3. Empirical validation

Three predictions of the theory are testable. First, *encoder ordering*: Sec. 5 confirms that the lowest- d/D encoder pairs achieve the largest radii and the highest- d/D pair (CLIP) the smallest. Second, *training vs. geometry*: Remark 1 predicts that the frozen geometry, not the training procedure, sets p_A ; indeed a no-noise baseline certifies within 1.9% of the noise-trained model (Table 2), because noise training can only refine the margin m inside $\mathcal{S}$, not change d/D . Third, *approximate invariance*: we measure the squared cosine alignment between PGD attack directions and $\mathcal{S}$ across the FakeAVCeleb test set and find $\cos^2 \theta \approx 0.18$. Only 18% of adversarial energy is on-manifold; the remaining 82% targets directions to which the classifier is (approximately) invariant. This near-orthogonality is precisely the regime in which Assumption 1 holds well enough for Theorem 1 to be predictive, and it directly quantifies the headroom that anisotropic smoothing exploits.

Table 1. Joint feature-space certification on FakeAVCeleb (five seeds, mean $\pm$ std). C@ r is certified accuracy (%) at radius r .

σ	Acc.	Abs.	Mean R	C@0.25	C@0.50	C@1.0/1.5
0.12	91.2 $\pm$ 1.0	0.3	0.288 $\pm$.003	88.8	0.0	0.0/0.0
0.25	91.2 $\pm$ 1.1	0.4	0.588 $\pm$.012	89.4	86.6	0.0/0.0
0.50	91.5 $\pm$ 1.1	0.4	1.165 $\pm$.034	90.5	89.2	85.4/0.0
1.00	92.5 $\pm$ 0.6	0.8	2.215 $\pm$.032	91.8	90.7	88.0/84.9

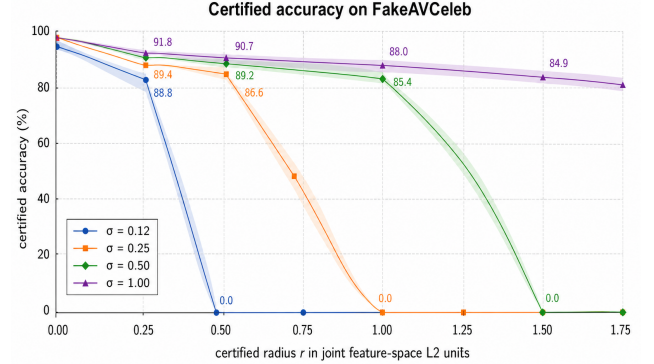


Figure 2. Certified accuracy vs. radius. Larger σ shifts the curve right; the $\sigma = 1.0$ model retains high certified accuracy through radius 1.5.

5. Experiments

Setup. FakeAVCeleb [12] is split deterministically into 3850 training, 825 validation, and 825 test clips, each with a 1:10 real/fake ratio (real = 0, fake = 1). Cross-dataset transfer uses 500 LAV-DF [3] clips. Models are trained with AdamW (learning rate 5×10^{-4} , weight decay 0.01), batch size 8 with gradient accumulation 4, cosine decay with 3 warmup epochs, gradient clipping 1.0, and early stopping. Main results average five seeds (42, 69, 420, 2026, 2804); Phase-2 studies (anisotropic, conformal) use seed 2026, justified by the low five-seed variance established below. Certification uses $n_0 = 100$, $n = 1000$, $\alpha = 0.001$.

5.1. Isotropic certification

Table 1 reports the headline isotropic result. Larger smoothing noise increases the certified radius *without* reducing clean accuracy, and abstention never exceeds 1% — both unusual for randomized smoothing and both explained by Theorem 1: the classifier sees only the on-manifold noise $\sigma\sqrt{d}$, so the margin is preserved while the radius grows with σ . The five-seed standard deviations are small ($\pm 0.6\%$ accuracy, ± 0.032 radius at $\sigma = 1.0$), which justifies single-seed Phase-2 studies. Figure 2 shows the full certified-accuracy curves.

Table 2. Baselines at $\sigma = 1.0$. The no-noise model certifies within 1.9% of the noise-trained model; feature-space PGD-AT inflates the radius but collapses validation AUC.

Model	Val AUC	EER	Acc.	Mean $\bar{R}$	C@1.0/1.5
Joint noise	0.941	0.139	92.5	2.215	88.0/84.9
No-noise	0.877	0.194	92.3	2.173	86.3/82.0
PGD-AT	0.730	0.335	90.5	2.463	90.5/90.5

Table 3. Five encoder families (no-noise baseline, $\sigma = 1.0$). d is the 90%-variance PCA dimension; $\mathcal{A} = \sqrt{D/d}$. Lower d/D tracks larger $\bar{R}$.

Encoder pair	D	d	d/D	$\mathcal{A}$	$\bar{R}$	Acc.
DINOv2-S + WavLM	1152	80	0.069	3.79	2.207	90.8
DINOv2-S + HuBERT	1152	79	0.069	3.82	2.173	90.2
DINOv2-B + Whisper-base	1280	119	0.093	3.28	2.132	92.2
DINOv2-S + Whisper-tiny	768	80	0.104	3.10	2.190	91.4
CLIP-B + Whisper-tiny	1152	126	0.109	3.02	1.948	94.3

5.2. Geometry, not training, drives certifiability

Table 2 isolates the role of training. The no-noise baseline reaches mean radius 2.173, only 1.9% below the noise-trained 2.215 — exactly as Remark 1 predicts, since the frozen geometry fixes d/D and only the margin can be tuned. Feature-space PGD adversarial training (PGD-AT) attains a larger radius (2.463) but its validation AUC collapses from 0.941 to 0.730 (EER 0.139 $\rightarrow$ 0.335); it trivially stabilizes predictions by flattening the decision boundary, certifying everything at the cost of discrimination. CertAV should thus not be read as an adversarial-training recipe; its value is to expose and certify a favorable geometry already present in frozen representations.

5.3. Encoder-scaling study

Table 3 and Figure 3 test Corollary 1 across five pairs combining DINOv2 [15], CLIP [16], Whisper [17], HuBERT [11], and WavLM [4]. The trend predicted by the theory holds at the extremes: the lowest- d/D pairs (DINOv2-S + WavLM/HuBERT, $d/D \approx 0.069$) achieve the largest radii ($\bar{R} \approx 2.2$), while CLIP-B + Whisper-tiny ($d/D = 0.109$) achieves the smallest (1.948) *despite* the highest clean accuracy (94.3%). The ordering is not strictly monotone in the mid-range, so d/D is the dominant but not the sole predictor — spectral decay rate and feature non-Gaussianity likely also contribute, and we therefore claim a dominant trend rather than an exact functional law. The accuracy/certifiability trade-off is itself useful guidance: contrastive encoders favor clean accuracy, self-supervised encoders favor certifiability.

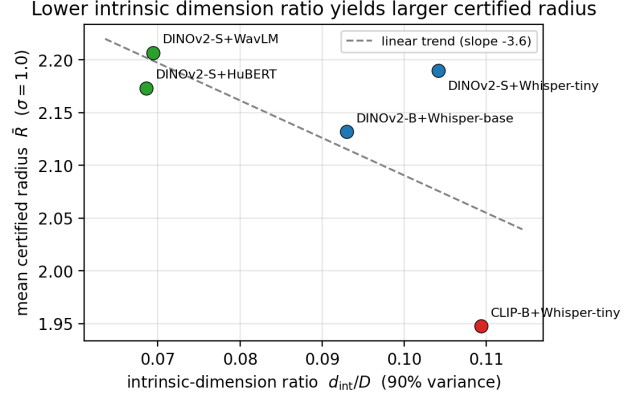


Figure 3. Encoder scaling. Self-supervised audio encoders (WavLM, HuBERT) give the lowest d/D and largest radii; the contrastive CLIP visual encoder gives the highest d/D and the smallest radius despite the best clean accuracy.

Table 4. Anisotropic smoothing at equalized budget trace(Σ) = $D\sigma^2$ ($\sigma = 1.0$, seed 2026). R_S is the on-manifold radius; R_{ℓ_2} the worst-case radius. For anisotropic rows, $C_S@1.0$ is certified accuracy w.r.t. the on-manifold radius R_S , *not* the worst-case R_{ℓ_2} (the two coincide for the isotropic row); it is therefore not a like-for-like comparison with the isotropic 88.0, but the appropriate metric under a manifold-aware threat model. Strategy 2 gives the largest R_S and zero abstention.

Strategy	Acc.	Abs.	R_{ℓ_2}	R_S	$C_S@1.0$
Isotropic	90.9	0.8	2.215	2.215	88.0
S1: eigenvalue-prop.	90.9	0.7	0.0002	6.272	89.5
S2: subspace proj.	90.9	0.0	0.0025	7.632	90.9
S3: inverse-eigenvalue	90.9	0.7	0.0139	6.586	89.0

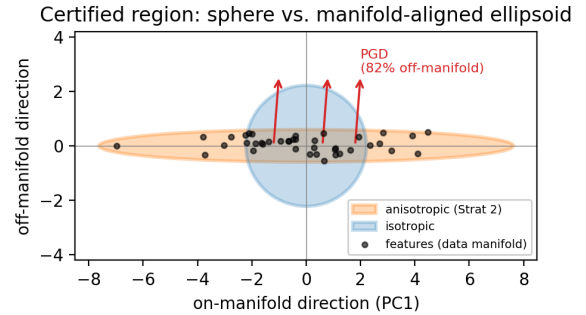


Figure 4. Isotropic smoothing certifies a sphere (radius 2.22, equal in all directions); manifold-aware smoothing (S2) certifies an ellipsoid with on-manifold axis 7.63 and near-zero off-manifold axes. PGD attacks point 82% off-manifold ($\cos^2 \theta \approx 0.18$), where the classifier is provably invariant.

5.4. Manifold-aware anisotropic smoothing

Table 4 reports the three strategies of Sec. 3.4 at *equalized* budget; the improvement is purely from redistribution.

Table 5. Split conformal prediction on the $\sigma = 1.0$ model ($\alpha = 0.10$, seed 2026). Marginal coverage and singleton rate are stable under PGD; class-conditional coverage exposes the 1:10 imbalance.

Setting	Cover.	Single.	Cov(real)	Cov(fake)
Clean	96.8	92.4	72.0	99.3
PGD $\varepsilon=0.25$	96.2	91.3	66.7	99.2
PGD $\varepsilon=0.50$	95.3	91.2	58.7	98.9
PGD $\varepsilon=1.00$	94.7	88.2	54.7	98.7

Strategy 2 (subspace projection) achieves an on-manifold radius of 7.63, a $3.4\times$ lift over isotropic (2.22), and attains 0% abstention — every correctly classified clip is certified, because concentrating noise inside $\mathcal{S}$ minimizes what the classifier head sees and maximizes p_A . The near-zero R_{ℓ_2} is *by design* (Fig. 4): the certificate is an ellipsoid with long axes in the 80 on-manifold directions and tiny axes in the 688 off-manifold directions where the classifier is invariant. Under a manifold-aware threat model — the relevant one, given $\cos^2 \theta \approx 0.18$ — the on-manifold radius is the correct adversarial metric, and an attacker spreading budget uniformly faces an even larger effective barrier since most energy lands off-manifold. This is the paper’s headline result.

5.5. Robust conformal prediction

Table 5 reports conformal prediction at the primary operating point $\alpha = 0.10$. Standard split conformal achieves target marginal coverage ($> 90\%$) on clean data and degrades gracefully under PGD (to 94.7% at $\varepsilon = 1.0$). The robust variant of Gendler et al. [10] restores guaranteed coverage by enlarging sets: at $r \geq 0.25$ the calibrated threshold reaches 0.5, so every set becomes {real, fake} — 100% coverage at 0% efficiency. This is the fundamental coverage-efficiency trade-off, and we therefore report $\alpha = 0.10$ as the operating point rather than the conservative $\alpha = 0.05$. Marginal coverage meets the $1 - \alpha$ target throughout, but conditional coverage varies sharply by class: fake clips (majority) are covered near-perfectly ($> 98\%$) while real clips (minority) are covered far less (54.7–72.0%). This deserves emphasis rather than a footnote, because the “real” class is the safety-critical one — under-covering it corresponds to false deepfake accusations of genuine media. The gap is a known property of *marginal* split conformal under class imbalance [18] and is intrinsic to the dataset’s 1:10 ratio, not to our use of it; the standard remedy is *class-conditional* (Mondrian) calibration, which enforces the coverage target within each class at the cost of wider “real” sets, and which we leave as the natural next step. We report marginal coverage here for comparability with prior work and flag the conditional gap explicitly. Crucially, even with this caveat, conformal prediction returns a covered set for the samples

Table 6. Modality ablation (five seeds). Unimodal certificates can match or exceed joint radii but protect a strictly smaller threat set.

Mode	σ	Acc.	Mean R	C@0.25	C@0.50
Joint	0.25	91.2	0.588	89.4	86.6
Visual-only	0.25	91.0	0.589	89.3	86.5
Audio-only	0.25	90.9	0.607	90.3	89.5
Joint	1.00	92.5	2.215	91.8	90.7
Visual-only	1.00	92.3	2.273	91.6	90.7
Audio-only	1.00	91.9	2.404	91.7	91.5

Table 7. Transfer and degradation at $\sigma = 1.0$. LAV-DF is cross-dataset; degradations are on FakeAVCeleb.

Setting	Acc.	Mean R	C@0.50	C@1.0
LAV-DF transfer	74.6	1.626	69.2	61.4
Social degradation	91.3	—	90.5	—
H.264 CRF 28	90.7	—	89.9	—
JPEG quality 75	91.6	—	90.5	—

on which CertAV abstains, which is the practical complement to the per-sample certificate.

5.6. Modality, transfer, and degradation

Table 6 shows that audio-only smoothing at $\sigma = 1.0$ has the largest mean radius (2.404), but it leaves the visual stream outside the certified set; the joint certificate is the more conservative statement when either modality may be manipulated. Table 7 reports cross-dataset and degradation robustness: on LAV-DF the certificate transfers at 74.6% accuracy and 61.4% certified accuracy at radius 1.0 — useful but imperfect, confirming dataset shift as a real limitation — while common media degradations (social re-encoding, H.264, JPEG) preserve $\geq 90\%$ certified accuracy at radius 0.5.

Composed input-space certificate. To address the feature-space scope head-on, we estimate the empirical feature-to-input Lipschitz constant from input-space attacks: the measured displacement-to-perturbation ratio is tightly concentrated at $L \approx 50$ across attack magnitudes. Composing R_{cert}/L maps the feature certificate to an *approximate* input-space radius of ≈ 0.12 , at which $\approx 90.7\%$ of clips remain correct and certified. This is an empirical, probabilistic bridge — not a worst-case formal Lipschitz proof — but it gives a concrete sense of the input-space protection the certificate implies, and the methodology composes directly with any future formal encoder bound.

6. av-robustbench

To make every reported number reproducible and to give the community a common interface for AV robust detection, we release `av-robustbench`, a Python library that

organizes the evaluation logic of this paper. It provides: (i) a detector-adapter interface so any frozen-encoder detector can be plugged in; (ii) attacks — PGD- ℓ_∞ and PGD- ℓ_2 with joint AV projection, plus Square-Attack and AutoAttack-style wrappers; (iii) randomized-smoothing certification (isotropic and PCA-aligned anisotropic); (iv) a degradation battery (social re-encoding, H.264, JPEG, audio corruptions); and (v) JSON “robustness cards” and a leaderboard schema. Following the RobustBench philosophy [7] but adapted to multimodal forensic threat models, its role here is reproducibility: every attack, certificate, degradation, and transfer result in this paper maps to a structured entry rather than an ad-hoc script, so the full study can be regenerated end to end.

7. Limitations

We highlight five concrete limitations with future directions. (1) *Feature-space scope*: CertAV certifies the frozen representation, not raw pixels or waveforms; composing the certificate with an encoder Lipschitz bound to obtain an end-to-end input guarantee is open. (2) *Single primary dataset*: results center on FakeAVCeleb with transfer to LAV-DF; broader evaluation across manipulation families and datasets is needed, and the 61.4% transfer certified accuracy shows headroom. (3) *Class imbalance*: the 1:10 real/fake ratio depresses conditional coverage on the real class (Table 5); class-conditional conformal calibration is a natural fix. (4) *Computational cost*: each certified clip needs $n = 1000$ forward passes, standard for smoothing but limiting for real-time use; caching frozen features mitigates but does not remove this. (5) *Subspace-invariance approximation*: Theorem 1 assumes exact invariance, whereas $\cos^2 \theta \approx 0.18 \neq 0$ indicates small residual off-manifold sensitivity; tightening the theory to bounded (rather than exact) invariance would make the amplification factor a guarantee rather than an idealized upper bound.

8. Conclusion

CertAV makes certified robustness practical for audio-visual deepfake detection by placing the guarantee at the level of frozen multimodal features. Its central finding is geometric and predictive: the intrinsic dimension of the representation — not the training procedure — governs certifiability (Theorem 1), which explains the large isotropic radii, predicts encoder behavior (Corollary 1), and motivates manifold-aware anisotropic smoothing for a $3.4\times$ on-manifold gain with zero abstention. Robust conformal prediction completes the picture with a coverage guarantee for every sample. Together these results turn a striking empirical observation into a coherent, certified story, and establish representation geometry as a first-class design axis for trustworthy multimodal forensics.

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